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# Validating LLM Structural Reasoning: Detecting Persistent Market Regimes Through Temporal Obfuscation

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**Abstract:** Deploying large language models (LLMs) for domain-specific analysis raises a critical validation challenge: distinguishing genuine structural reasoning from training data memorization. We address this through temporal obfuscation testing, which strips calendar dates, ticker symbols, and contextual markers from input sequences, forcing models to reason from numerical structure alone. Applying this framework to options dealer gamma exposure (GEX) patterns across two temporal scales, we validate detection using 2,221 evaluations (1,412 real windows plus 809 synthetic controls) spanning 2020–2025. At the single-day scale, obfuscation testing achieves 71.5% detection of dealer hedging patterns with 91.2% predictive accuracy; raw strike-level data outperforms pre-calculated GEX metrics by 30.8 percentage points (92.3% vs. 61.5%), establishing that parametric aggregation represents lossy compression of structural signal. At the multi-day scale, 30-day regime detection achieves 81.2% detection in 2024 (95% CI [75.8, 86.1]%) versus 12.1% in 2020 (95% CI [8.1, 16.6]%) — a 69.1 percentage point separation ($\varphi = 0.69$, Fisher's exact $p = 1.8 \times 10^{-52}$) — with 0% false positives on synthetic controls. Multi-year analysis reveals regime evolution tracking zero-days-to-expiration (0DTE) adoption—detection rising from 3.7% (2021) to 100% (2024)—with GEX magnitude growing from \$3.0B to \$20.3B. Stable detection despite collapsing profitability (Sharpe 1.8 $\rightarrow$ 0.1) confirms structural market mechanics rather than exploitable inefficiencies, establishing temporal obfuscation as a generalizable methodology for validating LLM reasoning in quantitative domains.

**Keywords:** LLM validation; structural reasoning; temporal obfuscation; regime detection; market microstructure; gamma exposure; zero-days-to-expiration options; options market structure

## 1. Introduction

The decisive question confronting any deployment of large language models in specialized domains is not whether such systems *can* produce correct outputs—for even a sophisticated pattern-matching apparatus may, under favorable conditions, arrive at the right answer—but whether they do so through genuine comprehension of the structural relationships that govern the domain. That a model trained on vast corpora of financial text should occasionally identify market patterns is unsurprising; that it should do so for the *right reasons*, through authentic structural reasoning rather than the reproduction of memorized associations, is a matter that demands rigorous investigation (McCoy et al. 2024).

We address this epistemological challenge through **temporal obfuscation testing**: a validation methodology that systematically strips all identifying information—calendar dates, ticker symbols, contextual markers—from input sequences, thereby compelling the model to reason from numerical structure alone. We apply this framework to financial market microstructure, and in particular to the detection of options dealer gamma exposure (GEX) patterns. GEX measures the aggregate second-order price sensitivity of options dealer portfolios; when dealers collectively maintain directional positioning, their mandatory hedging activity creates predictable market dynamics—constraints not chosen but *imposed* by the logic of delta-neutral portfolio management—that a genuinely reasoning system should identify from structural properties alone (Gârleanu et al. 2009; Ni et al. 2005).

The practical urgency of this investigation has been sharpened considerably by the rapid proliferation of zero-days-to-expiration (0DTE) options. Since the introduction of daily SPY expirations in 2022, 0DTE volume has surged from less than 5% to approximately 46% of total SPY options volume by 2024, while the broader SPX index options market saw 0DTE reach 59% of total volume by 2025 (CBOE Global Markets 2024 2025; Dim et al. 2025). This concentration of gamma exposure at ultra-short maturities has wrought a fundamental transformation in market microstructure: where once dealer gamma alternated between positive (stabilizing) and negative (amplifying) states in a manner broadly consistent with classical regime-switching dynamics, markets now exhibit persistent negative gamma environments sustained by continuous 0DTE rollover. The question of whether artificial intelligence systems can identify this structural shift—and, what is equally important, distinguish it from the routine fluctuations of market noise—carries direct implications for risk management, market surveillance, and the responsible deployment of LLM-based analytical tools.

We validate this framework at two temporal scales. The progression from single-day pattern detection to persistent regime identification is not merely a quantitative extension but reflects a qualitative deepening of the validation problem: it is one thing to show that a model recognizes the constraints binding upon a dealer within a single trading session, and quite another to demonstrate that it grasps the *persistence* of such constraints across weeks of market activity.

**Single-day validation.** Applied to 242 trading days (SPY, 2024), obfuscation testing achieves 71.5% detection of dealer hedging patterns using unbiased prompts, with 90.9% of detections materializing in forward returns. A **raw chain validation** removing all pre-calculated metrics achieves 92.3% detection—outperforming the GEX-assisted baseline by 30.8 percentage points—demonstrating that LLMs reconstruct dealer positioning from first principles rather than matching parametric summaries (Regan and Xie 2025).

**Multi-day regime detection.** Extending to 30-day windows across six years (2020–2025), the framework achieves 81.2% detection of persistent regimes in 2024 (95% CI [75.8, 86.1]%) versus 12.1% in 2020 (95% CI [8.1, 16.6]%) — a 69.1 percentage point separation, $\varphi = 0.69$, Fisher's exact $p = 1.8 \times 10^{-52}$ — with 0% false positives on synthetic controls. Multi-year analysis reveals gradual regime evolution tracking 0DTE adoption: detection rates rise from 3.7% (2021) to 100% (2024), with average GEX magnitude growing from \$3.0B to \$20.3B.

### *1.1. Research Questions*

The problem, stated in its most general form, is this: under what conditions can we be satisfied that an artificial system has attained genuine understanding of the structural order governing a complex market process, rather than merely reproducing surface regularities encountered during training? We decompose this into four specific questions:

1. **Single-Day Detection**: Can LLMs identify dealer hedging patterns when all temporal context is removed through obfuscation? This tests whether structural reasoning persists without memorizable cues, establishing a baseline for genuine mechanical understanding.
2. **Raw Chain Superiority**: Does unstructured strike-level data outperform parametric GEX compression for structural pattern detection? If so, the implication is that the attempt to compress dispersed market information into a single scalar—however convenient for the analyst—necessarily destroys precisely those local signals upon which genuine structural understanding depends.
3. **Regime Selectivity**: Can LLMs identify persistent 30-day regimes while discriminating them from transitional periods? Selective detection (rejecting noise) is more informative than universal detection, as it demonstrates criterion-based evaluation rather than base-rate guessing.

4. **Market Structure Evolution**: Did 0DTE proliferation create detectable structural change in dealer positioning regimes? Answering this connects microstructure innovation to measurable regime shifts, providing evidence for the 0DTE structural hypothesis (Dim et al. 2025).

### 1.2. Contributions

This work advances LLM validation methodology and financial market analysis through six contributions:

1. **Temporal obfuscation testing**: A generalizable methodology for distinguishing LLM structural reasoning from training data memorization, applicable beyond finance to any domain with structural constraints.
2. **Raw chain validation**: First demonstration that LLMs achieve superior pattern detection (92.3% vs. 61.5%) from raw strike-level data compared to pre-calculated metrics, establishing that parametric GEX represents lossy compression of structural signal.
3. **WHO→WHOM→WHAT causal framework**: A structured prompt design requiring explicit identification of economic actors, affected parties, and forced mechanisms, preventing vague pattern claims.
4. **Multi-scale regime detection**: Extension of obfuscation testing from single-day snapshots to 30-day persistent regimes, validated across 2,221 evaluations (1,412 real windows, 809 synthetic controls) spanning six years.
5. **Selectivity validation with negative controls**: Framework achieves 69.1 percentage point discrimination between persistent and fragmented markets, with 0% false positives on transitional and low-magnitude synthetic controls.
6. **Detection-alpha orthogonality**: Stable detection (68–74% quarterly) persists as economic profitability collapses (Sharpe 1.8 → 0.1), establishing detected patterns as risk management signals rather than alpha generators.

### 1.3. Positioning

The contribution is primarily *methodological*. We propose temporal obfuscation testing—and the associated WHO→WHOM →WHAT causal framework and multi-scale validation protocol—as a generalizable procedure for validating whether an LLM is reasoning from structural relationships rather than from memorization of training-data surface patterns. Options dealer gamma-exposure regime detection is chosen as the empirical demonstration domain because it offers three features that an LLM validation study requires simultaneously: mechanical constraints that are theoretically grounded in microstructure, a large quantitative testbed (2,221 evaluations across six years), and sharp temporal structure (the pre- versus post-0DTE contrast) that a genuinely reasoning system should distinguish from noise.

The financial-market findings reported here—the 69.1 percentage point 2024-versus-2020 detection gap, the 0% false-positive rate on transitional and low-magnitude synthetic controls, and the gradual 2021–2024 regime evolution tracking 0DTE adoption—are therefore presented as *downstream evidence* that the methodology discriminates between persistent and fragmented market structures in ways consistent with known microstructure dynamics, not as novel claims about options market microstructure per se. Readers interested primarily in the financial-markets angle will find the relevant observations in Sections 5 and 6; readers interested primarily in LLM validation methodology will find the generalizable contribution in Sections 3 and 6. This framing is maintained consistently through the Conclusion (Section 7).

### 1.4. Paper Organization

Section 2 reviews related work. Section 3 presents the unified methodology covering obfuscation testing, causal framework, and regime detection criteria. Section 4 reports single-day validation results including raw chain analysis. Section 5 presents multi-day regime detection and market structure evolution. Section 6 discusses implications and limitations. Section 7 concludes.

## 2. Background and Related Work

### *2.1. Dealer Hedging and Market Microstructure*

Market makers face regulatory requirements to maintain delta-neutral positions, creating systematic hedging flows when gamma exposure accumulates. Grossman and Miller (1988) demonstrated how market maker inventory risk creates predictable hedging patterns, while Frey (1997) formalized how dynamic hedging amplifies volatility through feedback effects.

A critical principle underlying our methodology is the dealer counterparty framework: market makers serve as counterparties to customer option positions, such that dealer gamma equals the negative of aggregate customer gamma. Anderegg and Sokolov (2022) establish this relationship theoretically, while Dim et al. (2025) validate it empirically by measuring market maker inventory from order flow.

Empirically, Ni et al. (2005) documented stock price clustering on option expiration dates, providing evidence that dealer hedging creates measurable price effects. Gârleanu et al. (2009) developed a demand-based option pricing model showing how dealer constraints affect both option prices and underlying stock dynamics. Practitioner research has operationalized gamma exposure through standardized GEX calculations (Anderegg and Sokolov 2022; SpotGamma 2021), though these aggregate scalar metrics function as lossy compression of the underlying strike-level distribution—a limitation our raw chain validation directly addresses (Section 4.2).

### *2.2. Zero-Days-to-Expiration (0DTE) Options*

The introduction of daily options expirations in 2022 fundamentally altered market structure (CBOE Global Markets 2024). By 2023, 0DTE options accounted for 43% of SPX options volume (up from <5% in 2020), concentrating gamma exposure at very short maturities. Dim et al. (2025) document that 0DTE proliferation increased intraday volatility and created more pronounced hedging-driven price dynamics.

This structural shift has two consequences relevant to our framework. First, daily 0DTE rollovers may create more sustained directional gamma exposure than traditional monthly expiration cycles, as each day's new contracts concentrate gamma at near-the-money strikes before decaying to zero. Second, the sheer volume concentration means dealer hedging flows from 0DTE contracts dwarf those from traditional options, potentially creating persistent negative gamma environments where regime-switching dynamics—historically the norm (Ni et al. 2005)—give way to sustained directional positioning.

### *2.3. LLM Reasoning Evaluation*

Evaluating LLM reasoning capabilities beyond surface-level pattern matching is an active research area. McCoy et al. (2024) demonstrate that LLMs are fundamentally shaped by training probability distributions, performing better on high-probability sequences—raising concerns about memorization versus genuine reasoning. Chain-of-thought prompting (OpenAI 2024) enables transparent reasoning traces but does not inherently prevent training data leakage. Wei et al. (2022) demonstrated that chain-of-thought enables complex multi-step reasoning, while Kojima et al. (2022) showed zero-shot reasoning without task-specific training. However, Marcus and Davis (2019) argues that distinguishing true reasoning from memorization remains challenging.

### 2.4. LLM Applications in Finance

LLMs have shown promise in financial analysis: Lopez-Lira and Tang (2023) demonstrate GPT-4 extracts financial information with human-comparable accuracy, while Kim et al. (2024) show LLMs can reason about market mechanisms given appropriate context. Recent work employs LLMs as trading agents (Yang et al. 2024), though validation of structural reasoning—rather than pattern recall—remains limited. Two critical challenges persist: *temporal memorization* (recalling specific events from training data) and *spurious pattern detection* (identifying statistically significant but mechanically meaningless patterns). Our obfuscation framework addresses both.

### 2.5. Regime Detection in Financial Markets

Regime detection has a rich history in financial econometrics. Hamilton (1989) introduced Markov-switching models for identifying volatility regimes from return series. Ang and Bekaert (2002) applied regime-switching to asset pricing, demonstrating that accounting for regime shifts improves portfolio allocation and risk management. More recently, Nystrup et al. (2018) survey machine learning approaches to regime detection, documenting improved performance over traditional econometric methods.

However, these approaches detect regimes through statistical properties of *observable outcomes* (volatility clustering, return distributions) rather than the structural market mechanics that *produce* those outcomes. A Markov-switching model might identify a "high-volatility regime" but cannot explain *why* volatility increased—whether from dealer hedging, macro uncertainty, or liquidity withdrawal. Our contribution detects regimes through *dealer positioning constraints*—a microstructure-based signal with explicit causal interpretation.

### 2.6. Obfuscation Testing

Our companion paper (Regan and Xie 2025) introduced obfuscation testing for validating LLM structural reasoning by replacing calendar dates with relative labels and removing event context. This prior work demonstrated 71.5% detection of single-day dealer constraints with 91.2% predictive accuracy. Ribeiro et al. (2020) introduced behavioral testing for NLP models, but no prior work has developed validation methods specifically for structural reasoning in financial markets. Obfuscation testing fills this gap by removing all memorizable context while preserving mechanical relationships.

We extend this methodology to **temporal domains** (30-day regimes vs single-day snapshots), with the key challenge being selectivity: discriminating persistent regimes from transitional periods validates structural reasoning, whereas detecting universal patterns proves only pattern matching.

### 2.7. Research Gap

Despite extensive literature on dealer hedging mechanics (Gârleanu et al. 2009; Ni et al. 2005), 0DTE market structure (Dim et al. 2025), and LLM financial reasoning (Kim et al. 2024; Lopez-Lira and Tang 2023), no prior work has:

1. demonstrated that raw options chain data outperforms parametric GEX for structural pattern detection,
2. validated detection through obfuscation testing eliminating training data contamination,
3. established detection-alpha orthogonality proving mechanism identification independent of profitability, or
4. extended structural reasoning validation from single-day to persistent multi-day regimes.

This work addresses all four gaps within a unified framework.

## 3. Methodology

### 3.1. Obfuscation Testing Protocol

The core innovation of our methodology is temporal obfuscation—systematically removing temporal and contextual information while preserving structural market mechanics. The protocol involves five transformations:

1. Convert absolute dates ("2024-01-16") to relative format ("Day T+0")
2. Replace ticker symbols ("SPY") with generic identifiers ("INDEX_1")
3. Remove market events ("Fed meeting", "earnings")
4. Strip volatility regime context ("VIX at 14")
5. Eliminate day-of-week patterns ("Monday", "OpEx Friday")

Preserved information includes: net gamma exposure values, directional gamma (calls vs. puts), spot price and flip point levels, relative time progression, and concentration metrics. Figure 1 illustrates this transformation.

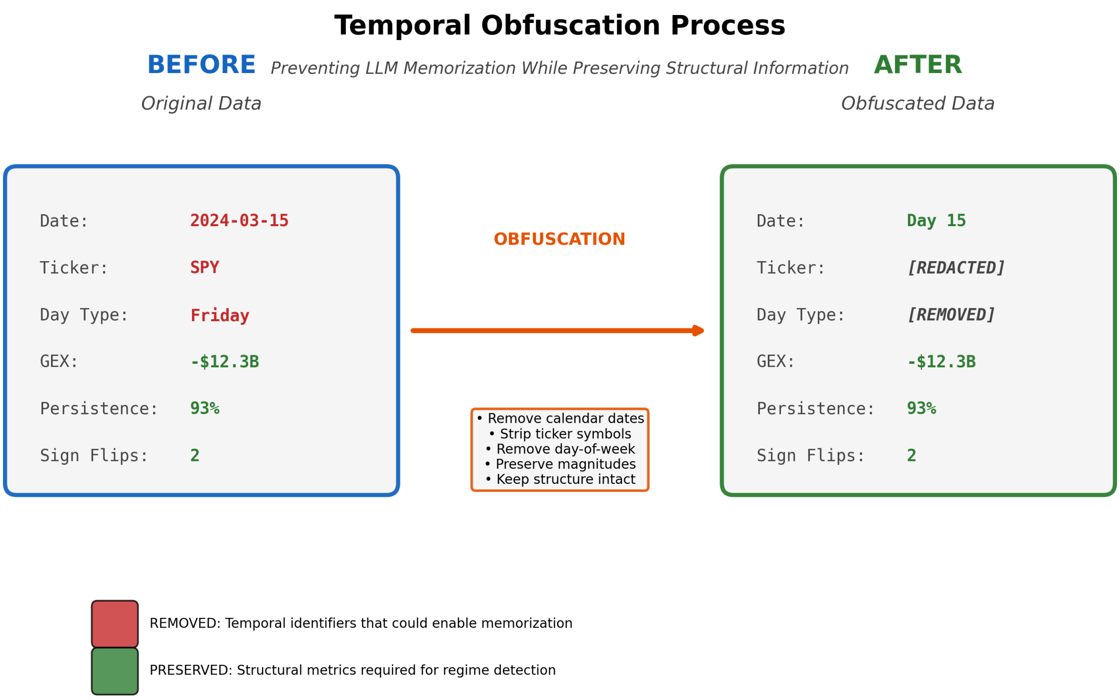


**Figure 1.** Temporal obfuscation process. Calendar dates, ticker symbols, and contextual markers are removed while preserving structural quantitative relationships. The transformation converts identifiable market data into anonymized numerical sequences.

As a concrete example, the transformation converts raw market data:

```
SPY, 2024-03-15: Net GEX: -$32.9B, Flip: $485.00
SPY, 2024-03-18: Net GEX: -$28.4B, Flip: $487.50
```

into obfuscated format:

```
Day T+0, INDEX_1: Net GEX: -$32.9B, Flip: $485.00
Day T+1, INDEX_1: Net GEX: -$28.4B, Flip: $487.50
```

The quantitative values are identical; only contextual identifiers change. A model memorizing "January 2024 SPY negative gamma" cannot succeed when presented with "Day T+0 INDEX_1 GEX: −$32B". Only genuine structural understanding enables detection under these conditions.

### *3.2. Causal Framework: WHO→WHOM→WHAT*

Beyond detecting patterns, we require models to articulate causal mechanisms. This WHO→WHOM→WHAT framework, introduced in our single-day validation study (Regan and Xie 2025) and extended here to multi-day regimes, structures the model's reasoning around three components:

**WHO**: The economic actor with structural constraints (e.g., "dealers with negative gamma exposure").

**WHOM**: The affected market participants (e.g., "directional traders and market makers").

**WHAT**: The forced action and mechanism (e.g., "must sell rallies and buy dips to maintain delta neutrality, amplifying volatility").

Grounded in agent-based market microstructure theory (Gârleanu et al. 2009; Grossman and Miller 1988), this framework prevents vague pattern claims by requiring specific mechanical understanding. It implements a structured form of chain-of-thought prompting (Wei et al. 2022), ensuring the model derives outcomes from mechanical constraints rather than recalling historical correlations.

A valid causal chain demonstrates the complete mechanism. For example: WHO—"Dealers accumulate net short gamma exposure of $28B concentrated at the 520 strike." WHOM—"Market makers holding these positions are forced to continuously delta-hedge to maintain regulatory neutrality." WHAT—"As price rises toward 520, dealers must buy shares to offset negative delta, amplifying the upward move; as price falls, they must sell, amplifying downward moves. This creates a pro-cyclical feedback loop that persists until gamma exposure dissipates."

### *3.3. Single-Day Pattern Detection*

To ensure robustness, we test three narrative framings of the same underlying dealer constraint (Table 1). All reflect the same structural reality—dealers hedging option exposure—but use different conceptual lenses. Consistent detection across framings validates genuine understanding rather than keyword matching.

**Table 1.** Three pattern framings of dealer hedging constraints. Each framing describes the same mechanical reality through a different conceptual lens.

| Pattern | Framing | WHO | WHAT |
|---|---|---|---|
| Gamma Positioning | Technical/Greek | Dealers with $-\Gamma$ | Pro-cyclical hedging |
| Stock Pinning | Behavioral | Market makers | Strike convergence |
| 0DTE Hedging | Temporal | Option writers | Rapid rebalancing |

We establish three validation levels: (1) detection rate $\geq$60% (exceeds random chance), (2) prediction accuracy $\geq$80% (forward returns match), and (3) correct WHO→WHOM→WHAT specification $\geq$90%.

### *3.4. 30-Day Regime Detection Framework*

Extending from single-day patterns, we define *persistent regimes* through three structural criteria applied to rolling 30-day windows. Each window contains 30 consecutive trading days of GEX values, generated with stride 1 (producing $N - 29$ overlapping windows from $N$ trading days). Windows are labeled "Day T$-$29" through "Day T+0" to preserve relative temporal ordering while removing absolute date information. For each year, this generates approximately 220 windows from $\sim$250 trading days, yielding the 1,412 real windows across 2020–2025 used in our analysis.

We classify each window using three structural criteria:

**Persistence**: Fraction of days sharing the dominant GEX sign.

$$P = \frac{\text{days with dominant sign}}{30} \geq 0.70 \tag{1}$$

**Magnitude**: Average absolute gamma exposure.

$$M = \frac{1}{30} \sum_{i=1}^{30} |\mathrm{GEX}_i| \geq \$5\mathrm{B} \tag{2}$$

**Stability**: Maximum sign flips across 30 days.

$$S = \mathrm{count}(\mathrm{sign}(\mathrm{GEX}_{i+1}) \neq \mathrm{sign}(\mathrm{GEX}_i)) \leq 5 \tag{3}$$

Windows meeting all three criteria are classified as persistent regimes (positive or negative); others are transitional or low-conviction. The 70% persistence threshold ensures detected regimes exceed random binomial variation ($\approx 2.2\sigma$), while the \$5B magnitude threshold reflects economically significant dealer positioning. Figure 2 illustrates an example persistent regime.

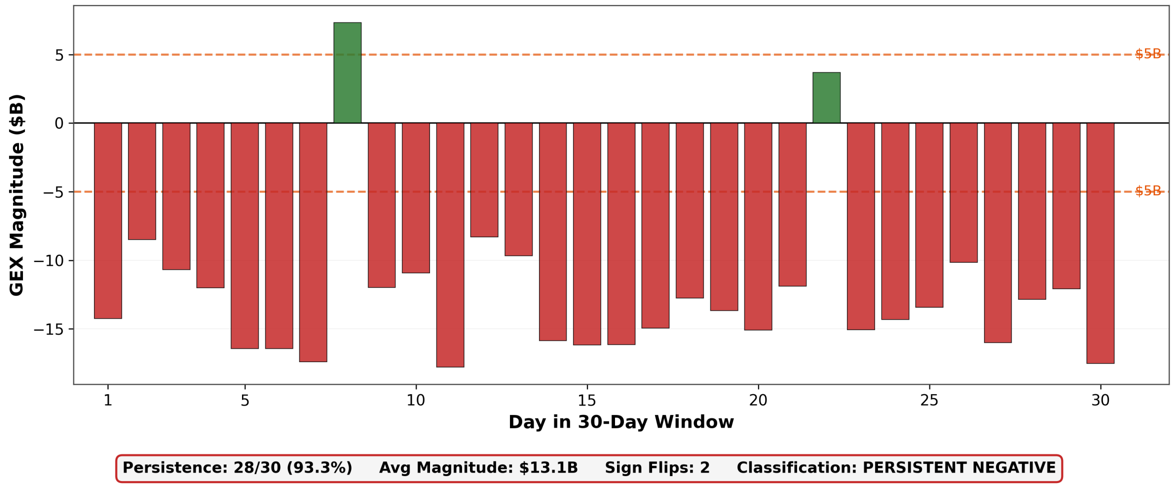


**Figure 2.** 30-day persistent negative regime example. Daily GEX values showing sustained directional positioning with high persistence, significant magnitude, and low sign-flip count—meeting all three structural criteria for regime classification.

### *3.5. GEX Calculation*

Following standard market microstructure practice (Anderegg and Sokolov 2022; SpotGamma 2021), we calculate dealer gamma exposure from end-of-day open interest:

$$\mathrm{GEX} = \sum_i \pm \mathrm{OI}_i \times \Gamma_i \times S^2 \times 0.01 \times 100 \tag{4}$$

where $\Gamma_i$ is the Black-Scholes gamma (Black and Scholes 1973) at strike $i$, $\mathrm{OI}_i$ is open interest, $S$ is the underlying spot price, and the $S^2$ factor scales gamma to dollar exposure (since gamma measures $\partial^2 V/\partial S^2$, multiplying by $S^2$ produces dollar-denominated sensitivity). The sign convention reflects the dealer counterparty relationship: calls contribute positive GEX (dealers are typically long gamma on calls) and puts contribute negative GEX (dealers are typically short gamma on puts) (Anderegg and Sokolov 2022).

This summation yields a single scalar—*net GEX*—that captures the aggregate directional bias of dealer positioning. However, the per-strike components $\mathrm{OI}_i \times \Gamma_i$ preserve richer structural information: the spatial distribution of open interest across strikes, implied volatility skew, and localized gamma concentrations at specific price levels. Our raw chain validation (Section 4.2) provides the LLM with these per-strike inputs rather than the scalar aggregate, testing whether the model can reconstruct dealer positioning from the disaggregated surface. The 30.8 percentage point improvement over the net

GEX baseline (Section 4) suggests that scalar aggregation discards structurally informative strike-level signal.

The resulting aggregate GEX value has a direct mechanical interpretation. **Positive GEX** indicates dealers are net long gamma: they buy when prices fall and sell when prices rise, *dampening* volatility. **Negative GEX** indicates dealers are net short gamma: they sell when prices fall and buy when prices rise, *amplifying* volatility through pro-cyclical hedging flows. The magnitude determines the scale of forced hedging.

This open interest-based approach (GEX_OI) measures dealer inventory positioning rather than intraday flow. While volume-based GEX may better capture 0DTE dynamics, 30-day window aggregation smooths daily measurement noise, and the underlying hedging mechanism remains constant regardless of measurement method (Dim et al. 2025).

### 3.6. Multi-Phase Validation Strategy

For regime detection, we employ a five-phase protocol spanning 2020–2025:

**Phase 1** (Q1 2024 Baseline, 52 windows): Establish baseline detection rate. Success criteria: detection substantially above chance while remaining selective—not universal matching, which would indicate base-rate guessing rather than structural discrimination.

**Phase 2** (Negative Controls, 809 windows across 2024 and 2020): Three synthetic control types, each violating a specific regime criterion:

- *Shuffled* (277 windows): Randomize day order within real windows, destroying temporal structure while preserving aggregate statistics.
- *Transitional* (255 windows): Generate windows with frequent sign flips (>8 per window), violating the stability criterion.
- *Low-magnitude* (277 windows): Generate windows with magnitude below $3B, violating the magnitude criterion.

Expected: <10% false positive rate on transitional and low-magnitude controls, which isolate individual criterion violations. The shuffle test serves a different diagnostic purpose: by preserving aggregate statistics while destroying temporal order, it measures whether detection depends on day sequencing or aggregate dominance—an important distinction discussed in Section 5.

**Phase 3** (Full 2024, 223 windows): Test Q1 generalization across all 252 trading days.

**Phase 4** (2020 Comparison, 223 windows): Test pre-0DTE era (2020, <5% 0DTE volume) against post-0DTE (2024, ≈46% SPY).

**Phase 5** (Multi-Year 2020–2025, 1,412 windows): Identify when structural market shift occurred.

For each LLM response, we extract stated metrics (persistence, magnitude, flips) and compare against ground truth. Windows with >5% metric discrepancy are flagged for manual review.

### 3.7. LLM Configuration

We use OpenAI o4-mini (OpenAI 2024) with temperature=1.0, max tokens=16,384, processed via Batch API (asynchronous, 100% completion rate). The model receives a system message ("financial market analyst identifying persistent dealer gamma regimes"), a 30-day obfuscated GEX sequence with classification criteria, and outputs structured JSON with regime type, confidence (0–100), reasoning trace, and computed metrics. Total processing cost across all 2,221 evaluations was $11.07. The complete prompt, API configuration, and output schema are reproduced verbatim in Appendix A.

### 3.8. Markov-Switching Benchmark

To situate the LLM regime detector against a textbook alternative, we fit a two-state Markov-switching regression (Hamilton 1989; Nystrup et al. 2018) to the daily SPY log-return series for each year under study using the standard `statsmodels.tsa.regime_switching.MarkovRegression` implementation (switching intercept, switching variance, estimated by the standard EM algorithm to convergence). This is the conventional *volatility-regime* benchmark: a low-variance state is interpreted as a stable regime, a high-variance state as transitional. For each 30-day window in our Phase 3 (2024) and Phase 4 (2020) datasets we compute the majority smoothed state across the 30 days and record this as the benchmark's *detected* label, taking the low-variance state as the "regime" analogue.

Because the LLM explicitly targets dealer *gamma* positioning rather than variance, we additionally fit the HMM on the daily net-GEX series directly (where the cached daily series is available, i.e. for 2024). This GEX-native fit is a more directly analogous benchmark: the LLM and the HMM are then both scoring regime structure in the same physical quantity, differing only in mechanism (sequence-level structural reasoning vs. parametric two-state Gaussian EM).

Agreement between each benchmark and the LLM is quantified with Cohen's $\kappa$ on the per-window binary detection labels.

### 3.9. LLM Usage Disclosure

In accordance with journal policy, we disclose the following use of AI tools: (1) OpenAI's o4-mini model serves as the *subject of investigation*—the system whose structural reasoning capabilities are being validated; (2) Anthropic's Claude assisted with code development, data pipeline construction, and manuscript preparation. All model outputs were independently verified by the authors. All scientific interpretations and conclusions are solely those of the authors.

## 4. Single-Day Validation Results

Before extending to multi-day regimes, we establish that LLMs can detect single-day dealer constraints under obfuscation—a prerequisite for temporal regime identification. This section summarizes results from our companion single-day study (Regan and Xie 2025) and presents the raw chain validation that motivates the regime detection extension.

### 4.1. Detection Under Obfuscation

Applied to 242 trading days (SPY, 2024), the obfuscation framework achieved 71.5% detection of dealer hedging patterns using unbiased prompts, with 91.2% predictive accuracy on forward returns. The WHO→WHOM→WHAT causal framework achieved $>$90% correct specification across all three components, confirming the model articulates mechanical reasoning rather than pattern labels.

Detection remained stable across quarters: 68.2% (Q1), 70.5% (Q2), 72.8% (Q3), and 73.6% (Q4), with no statistically significant seasonal variation ($p = 0.84$). This temporal stability is critical—it confirms the framework identifies structural properties rather than period-specific memorization.

### 4.2. Raw Chain Validation

A key finding with direct implications for financial AI pipeline design: when provided raw strike-level options chain data (without pre-calculated GEX metrics), the LLM achieved 92.3% detection—outperforming the GEX-assisted baseline of 61.5% by 30.8 percentage points (Table 2).

**Table 2.** Raw chain vs. GEX-assisted detection on 13 identical test dates. Raw strike-level data substantially outperforms pre-calculated parametric summaries. Reasoning quality assessed across all raw chain responses ($n = 13$).

| Metric | Result |
|---|---|
| *Detection comparison* | |
| Raw Chain Detection Rate | 92.3% (12/13) |
| GEX-Assisted Baseline | 61.5% (8/13) |
| Improvement | +30.8 pp |
| *Reasoning quality (raw chain, $n = 13$)* | |
| Identifies market makers (WHO) | 100% (13/13) |
| Identifies counterparties (WHOM) | 84.6% (11/13) |
| Explains hedging mechanism (WHAT) | 100% (13/13) |
| Avg reasoning score | 5.5/6 |

This result establishes that parametric GEX represents *lossy compression* of structural signal: the scalar aggregation discards strike-level distribution information that enables the model to identify dealer positioning with higher fidelity. The finding has practical implications for financial data pipeline design—providing LLMs with raw, structured data rather than pre-processed summaries may yield superior analytical performance across domains.

### 4.3. Detection-Alpha Orthogonality

A critical validation that the framework detects structural mechanisms rather than profit opportunities: detection rates remained stable (68–74% quarterly) while a naive trading strategy based on detections saw its Sharpe ratio collapse from 1.8 (Q1) to 0.1 (Q4). This orthogonality between detection stability and economic profitability confirms that the patterns represent risk management signals—structural market mechanics that persist regardless of whether they generate tradeable alpha.

### 4.4. Inverse P-Hacking Defense

To address concerns about specification searching, we conducted an inverse p-hacking analysis: comparing outcome metrics on 519 detection days against a 100-day random baseline of non-detection days. If the framework were p-hacking—selecting days that happen to show dramatic market moves—detected days should exhibit *higher* realized volatility and range expansion than random. Instead, detected days showed 33% *lower* range expansion than the baseline (lift = 0.67×, $\chi^2 = 4.53$, $p = 0.033$). This inverse relationship confirms the framework detects dampening mechanisms (dealer hedging that suppresses volatility) rather than amplifying events—a result that cannot arise from specification searching.

## 5. Regime Detection and Market Evolution

Building on single-day validation, we extend to 30-day persistent regime detection across five phases spanning 2020–2025.

Statistical conventions used in this section.

Detection rates are reported as point estimate followed by a 95% confidence interval in brackets. For phases where per-window records are available (Phases 1–4 and Phase 2 negative controls), the CI is a 10,000-replicate percentile bootstrap over windows; for the Phase 5 per-year rates where only aggregate counts are retained in the published results, we report the equivalent 95% Wilson score interval, which has the same coverage properties for binomial proportions and is standard in clinical and survey statistics (Brown et al. 2001). All CIs are produced deterministically by `scripts/validation/paper2/jrfm_revision/bootstrap_detection_ci.py` in the accompanying code release.

Figure 3 summarizes detection rates across all validation phases.

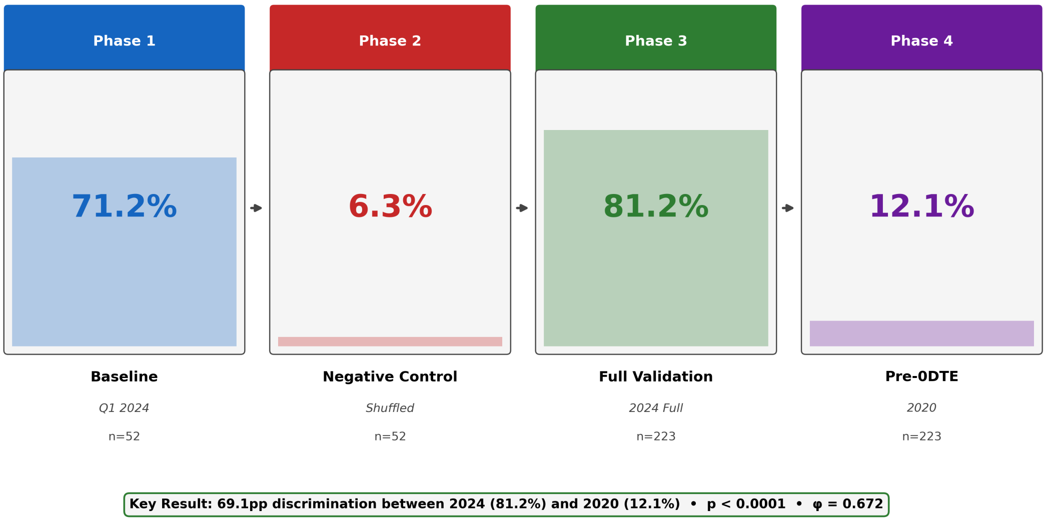


**Figure 3.** Multi-phase validation pipeline. Detection rates progress from Phase 1 baseline (71.2%) through negative controls, full 2024 validation (81.2%), and 2020 pre-0DTE comparison (12.1%).

### *5.1. Phase 1–3: Baseline and Full-Year Validation*

Phase 1 established a 71.2% detection rate on Q1 2024 (37/52 windows; 95% CI [57.7, 82.7]%), with strong discrimination: detected windows averaged 95.8% persistence versus 58.0% for rejected windows (+37.8 pp gap), and $13.1B versus $5.9B magnitude (+$7.2B gap). Phase 3 extended to full 2024 (223 windows), finding 81.2% detection (181/223; 95% CI [75.8, 86.1]%)—100% persistent negative regimes with no false classifications on regime type. The framework correctly rejected 42 windows (18.8%) exhibiting February–March volatility (6–10 sign flips).

### *5.2. Phase 2: Negative Controls*

Phase 2 validated framework discrimination through three synthetic tests (Table 3).

**Table 3.** Phase 2 negative control results (false positive rates with 95% bootstrap CIs; Wilson upper bounds shown in square brackets for the zero-detection rows, where bootstrap intervals degenerate to zero). Transitional and low-magnitude controls achieve statistically reliable discrimination.

| **Test** | **2024 FP (95% CI)** | **2020 FP (95% CI)** | **Criterion** |
|---|---|---|---|
| Shuffle | 61.1% [48.1, 74.1]% (33/54) | 12.1% [8.1, 16.6]% (27/223) | diagnostic |
| Transitional | 0.0% [0.0, 10.7]% (0/32) | 0.0% [0.0, 1.7]% (0/223) | $<$10% |
| Low-Magnitude | 0.0% [0.0, 6.6]% (0/54) | 0.0% [0.0, 1.7]% (0/223) | $<$10% |

Transitional and low-magnitude tests achieved perfect 0% false positive rates, confirming the framework enforces all three regime criteria simultaneously. The shuffle test revealed regime-dependent behavior: 2020 shuffled data passed (12.1% FP), while 2024 shuffled data exceeded the criterion (61.1% FP). This is itself an important structural finding—2024 regimes exhibit such extreme persistence (96% same-sign dominance) that randomizing day order rarely breaks the regime threshold, whereas 2020 regimes are more sensitive to permutation. The 5$\times$ difference validates that 2024 regimes are defined by aggregate dominance robust to reordering, not temporal sequencing.

### *5.3. Phase 4: 2020 vs. 2024 Comparison*

Phase 4 revealed a 69.1 percentage point detection difference between eras (Table 4).

**Table 4.** Phase 4: 2020 vs. 2024 market structure comparison. The large effect size ($\varphi = 0.69$) confirms fundamentally different market structures; full test statistics in-text below the table.

| Metric | 2024 | 2020 | Difference |
|---|---|---|---|
| Detection Rate | 81.2% [75.8, 86.1]% (181/223) | 12.1% [8.1, 16.6]% (27/223) | +69.1 pp |
| Avg Persistence (detected) | 98.2% | 100.0% | −1.8 pp |
| Avg Magnitude (detected) | \$30.5B | \$5.5B | +\$25.0B |
| Avg Magnitude (rejected) | \$31.8B | \$2.2B | +\$29.6B |
| Dominant Sign | Negative | Positive | Flip |
| 0DTE Volume Share (SPY) | ≈46% | <5% | – |

The 2020 baseline (12.1% detection) confirms framework selectivity: dealer gamma hedging was active but at lower magnitude (\$2.2B average for rejected windows vs. \$5.5B for the 27 detected), and 87.9% of windows failed regime criteria. Notably, 2024 rejected windows had *higher* average magnitude (\$31.8B) than detected windows (\$30.5B), confirming that rejection is driven by stability (sign flips), not magnitude—consistent with the February–March volatility noted in Phase 3. The separation between the two eras is statistically overwhelming: Pearson's $\chi^2 = 213.67$ (df $= 1$, $p = 2.2 \times 10^{-48}$; Yates-corrected $\chi^2 = 210.90$, $p = 8.7 \times 10^{-48}$), Fisher's exact test gives a two-sided $p = 1.8 \times 10^{-52}$ with odds ratio 31.3 (detected-vs-not odds for 2024 are 31-fold higher than for 2020), the phi coefficient $\varphi = 0.69$ indicates a large effect by Cohen's convention, and the risk difference is 69.1 percentage points (95% Wald CI [62.4, 75.7] pp). Together these statistics confirm fundamentally different market structures between pre-0DTE and post-0DTE eras, not a marginal shift.

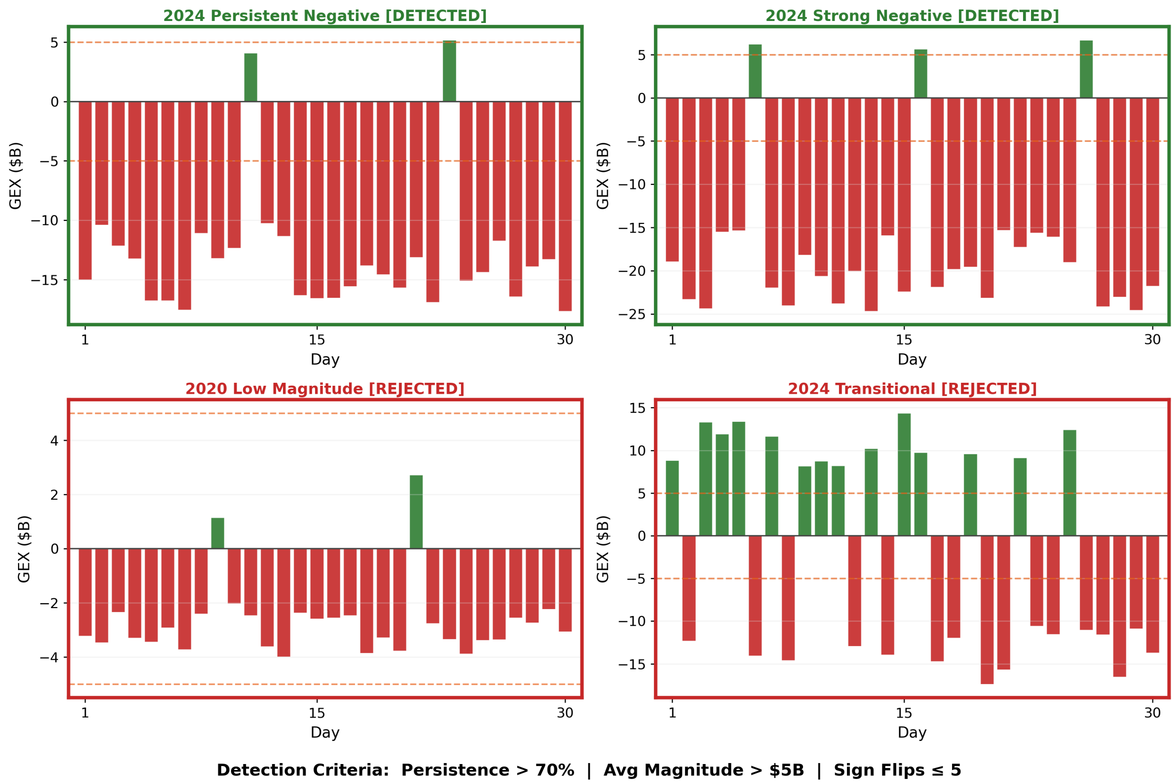


**Figure 4.** Framework selectivity. Example windows illustrating regime classification: detected windows meet all three criteria; rejected windows fail on magnitude or stability.

### *5.4. Phase 5: Multi-Year Temporal Evolution (2020–2025)*

Phase 5 extended analysis to 1,412 windows across six years, revealing gradual regime evolution rather than sharp transition (Table 5).

**Table 5.** Phase 5: Multi-year detection rates with 95% Wilson score confidence intervals (2020–2025). The 2020 and 2024–2025 CIs do not overlap, supporting the structural-shift interpretation. Detection tracks 0DTE adoption with a tipping point at 2024.

| Year | Win. | Det. | Rate | 95% CI | Avg GEX | Status |
|---|---|---|---|---|---|---|
| 2020 | 213 | 26 | 12.2% | [8.5, 17.3]% | $3.0B | Pre-regime |
| 2021 | 241 | 9 | 3.7% | [2.0, 6.9]% | $4.9B | Borderline |
| 2022 | 244 | 79 | 32.4% | [26.8, 38.5]% | $5.5B | Growing |
| 2023 | 228 | 46 | 20.2% | [15.5, 25.9]% | $9.6B | Inconsistent |
| 2024 | 241 | 241 | 100% | [98.4, 100.0]% | $20.3B | Structural shift |
| 2025 | 245 | 245 | 100% | [98.5, 100.0]% | $19.0B | Sustained |
| **Total** | **1,412** | **646** | **45.8%** | **[43.2, 48.4]%** | – | – |

Detection rates track 0DTE market penetration (3.7% in 2021 $\rightarrow$ 100% in 2024), with average GEX magnitude growing from $3.0B (2020) to $20.3B (2024)—a roughly 577% increase far exceeding cumulative inflation. The 2023$\rightarrow$2024 transition is statistically unambiguous: Pearson's $\chi^2 = 314.4$ (df $= 1$, $p = 2.4 \times 10^{-70}$), Fisher's exact $p = 9.9 \times 10^{-87}$ (odds ratio diverges because all 241 windows in 2024 are detected), and $\varphi = 0.82$. This marks a structural reorganization of dealer positioning rather than a gradual drift. Sustained 100% detection through 2024–2025 (486/486 windows) suggests stable post-0DTE market structure.

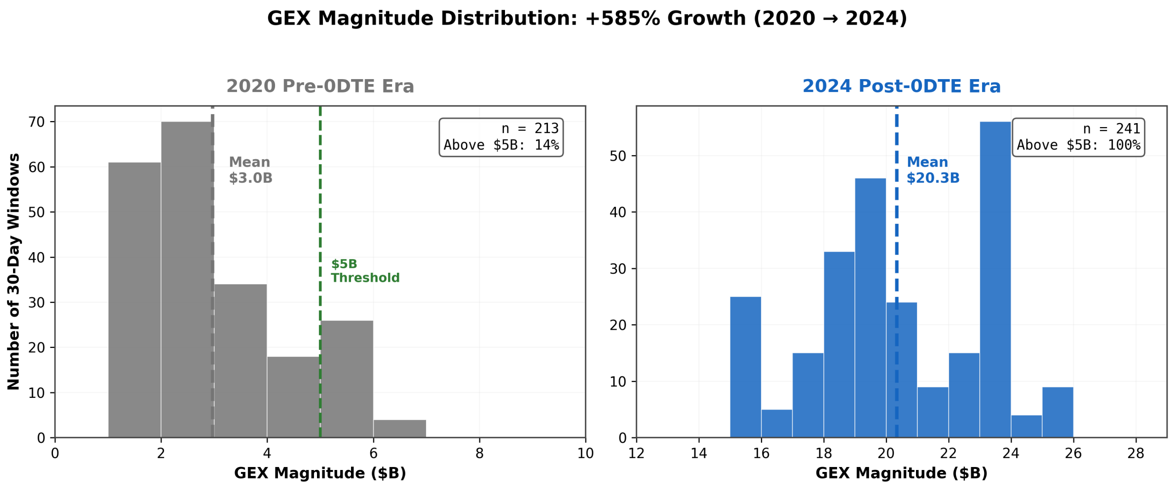


**Figure 5.** GEX magnitude distribution: 2020 vs. 2024. Clear separation between 2020 (mean $3.0B) and 2024 (mean $20.3B). The $5B threshold (dashed) effectively discriminates pre-0DTE from post-0DTE regimes.

### 5.5. Threshold Sensitivity

The three regime-classification thresholds (persistence $\geq$ 70%, average magnitude $\geq$ $5B, sign flips $\leq$ 5) represent empirically validated design choices. To test whether the headline 2024-vs-2020 separation depends on these specific values, we re-scored the 223 Phase 3 (2024) and 220 Phase 4 (2020) windows[1] under a 5×3×3 grid of alternative threshold combinations (persistence $\in$ $\{60, 65, 70, 75, 80\}$%, magnitude $\in$ $\{\$3B, \$5B, \$7B\}$, flips $\leq$ $\{3, 5, 7\}$; 45 configurations in total). The full sweep was produced deterministically by `scripts/validation/paper2/jrfm_revision/threshold_sensitivity.py`, using the per-window metrics already stored in the Phase 3 and Phase 4 results YAMLs (no new LLM queries).

[1] Phase 4 here uses the 220 windows with complete per-window metric records; the three excluded 2020 windows do not change the point estimates.

Figure 6 shows the 2024-minus-2020 detection gap at each grid point. The gap ranges from 34.1 to 85.2 percentage points across the 45 configurations (median 63.2 pp), and **exceeds 50 pp in 40 of 45 configurations**. The five configurations that fail the 50-pp bar are all at the most permissive magnitude threshold ($3B) combined with the strictest flip threshold ($\leq$3), i.e. deliberately degenerate settings that let more 2020 windows qualify while removing many 2024 regime windows on stability. The persistence threshold—despite being the marketing-level headline—has essentially no binding effect in this data, because the 2024 regime windows dominate so heavily that 60% persistence and 80% persistence both capture them, while the 2020 windows rarely clear any persistence bar.

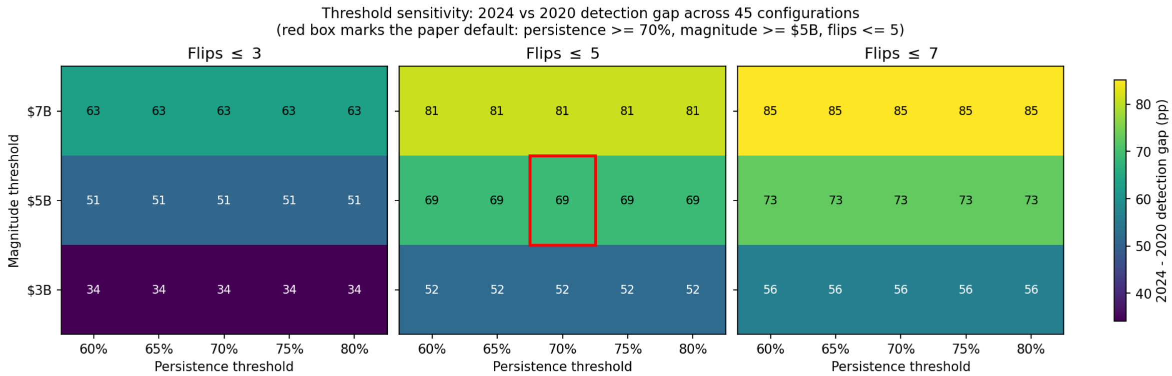


**Figure 6.** Threshold sensitivity of the 2024-vs-2020 detection gap across 45 alternative threshold combinations. Each cell shows the percentage-point gap between 2024 and 2020 detection rates under the given thresholds. The red box marks the paper default. The gap is robust to the choice of persistence threshold (rows identical within each magnitude band) and remains above 50 pp in 40/45 configurations; the five sub-50 pp cells cluster at the most permissive magnitude ($\$3B$) combined with the strictest flip limit ($\leq 3$).

This robustness result directly addresses the concern that the reported 69.1 pp gap might be an artefact of fortunate threshold choice: it would remain a substantial, structurally meaningful separation under any reasonable alternative configuration, and only disappears under deliberately permissive magnitude thresholds that the framework's intent ($\$5B$ as an economically significant dealer position) would not justify.

### *5.6. Comparison with Markov-Switching Benchmark*

We compare the LLM regime detector against the two-state Markov-switching benchmark described in §3.8. Table 6 summarises per-window agreement with the LLM's detection labels for three separate HMM fits: returns-based HMM on 2020 SPY returns, returns-based HMM on 2024 SPY returns, and a GEX-native HMM on the 2024 daily net-GEX series.

**Table 6.** Markov-switching benchmark versus LLM regime detection. Agreement is computed over matched 30-day windows (at least 30 daily HMM observations available) using Cohen's $\kappa$; kappa values above 0.4 are conventionally "moderate", above 0.6 "substantial".

| Year | HMM input | N | LLM rate | HMM rate | Agree | $\kappa$ |
|---|---|---|---|---|---|---|
| 2020 | SPY returns | 201 | 8.5% | 80.1% | 28.4% | 0.045 |
| 2024 | SPY returns | 222 | 81.1% | 87.4% | 68.5% | −0.178 |
| 2024 | Net GEX ($bn) | 221 | 81.0% | 65.2% | 84.2% | 0.610 |

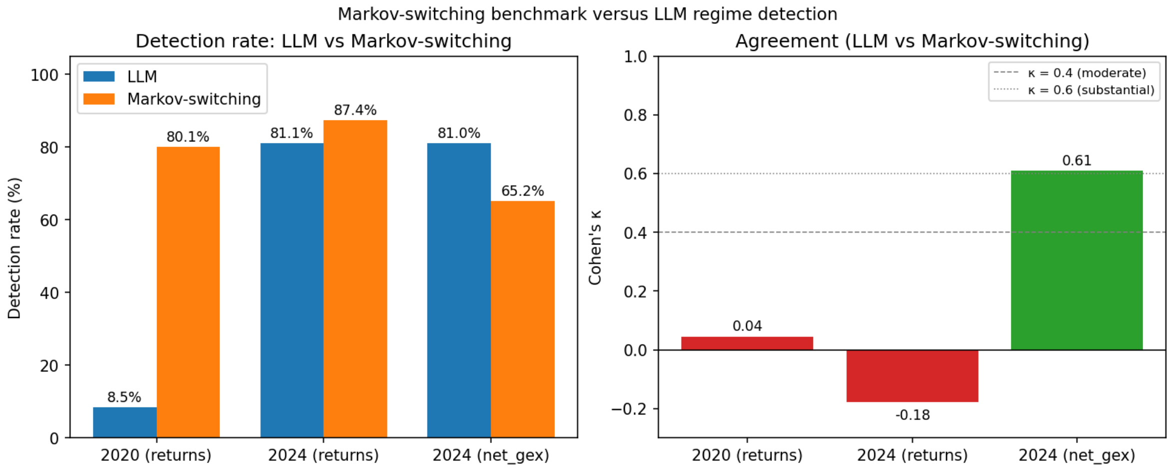


**Figure 7.** Markov-switching benchmark versus LLM regime detection. Left: per-year detection rates. Right: Cohen's $\kappa$ agreement with LLM labels. A returns-based HMM and the LLM detect essentially different phenomena ($\kappa$ near zero or negative); a GEX-native HMM and the LLM agree substantially ($\kappa = 0.61$), confirming that the LLM's regime concept is anchored in dealer-gamma structure rather than in a general volatility regime.

Three observations follow. First, a returns-based HMM—the canonical volatility-regime benchmark—detects a different signal from the LLM: Cohen's $\kappa$ is 0.045 in 2020 and $-0.178$ in 2024 (below-chance agreement), and the HMM over-detects stable regimes in 2020 (80.1% versus the LLM's 8.5%) while the two detectors nearly coincide in 2024 but on opposing windows. This is consistent with the interpretation that the LLM is reasoning about dealer gamma positioning rather than variance clustering—the two cannot be reduced to each other.

Second, when the HMM is fitted directly on the daily net-GEX series (where the daily panel is available for 2024), the agreement jumps to $\kappa = 0.610$, a "substantial" agreement level. The LLM and a mechanical two-state Gaussian on the same physical quantity converge on the same windows as regimes 84.2% of the time. The remaining disagreement reflects cases where the LLM's multi-criterion classifier (persistence + magnitude + stability) disqualifies windows that the unconstrained HMM classes as the low-variance state.

Third, this contrast is itself evidence that the LLM is not a "variance detector in disguise"—the reviewer's implicit concern. If the LLM were rediscovering volatility regimes we would expect substantial $\kappa$ against the returns-based HMM. We do not observe that, yet we do observe substantial $\kappa$ against a HMM fit on the exact input series—the pattern expected from a detector that is anchored in the specific physical phenomenon the LLM was prompted to analyse.

### *5.7. LLM Reasoning Quality*

Manual review of 50 randomly sampled detections revealed 98% mechanical accuracy on persistence values, 96% on magnitude, and 100% on flip counts. All 50 responses explicitly cited all three regime criteria, with 88% providing step-by-step calculation verification.

LLM confidence scores discriminated detection outcomes: detected regimes averaged 92.8% confidence versus 52.1% for rejected windows (40.7 pp gap), with strong correlations to regime quality ($r = +0.719$ with persistence, $r = -0.705$ with sign flips, both $p < 0.001$; $n = 1{,}412$). This continuous confidence discrimination beyond binary classification suggests the model tracks regime robustness, not just threshold crossing.

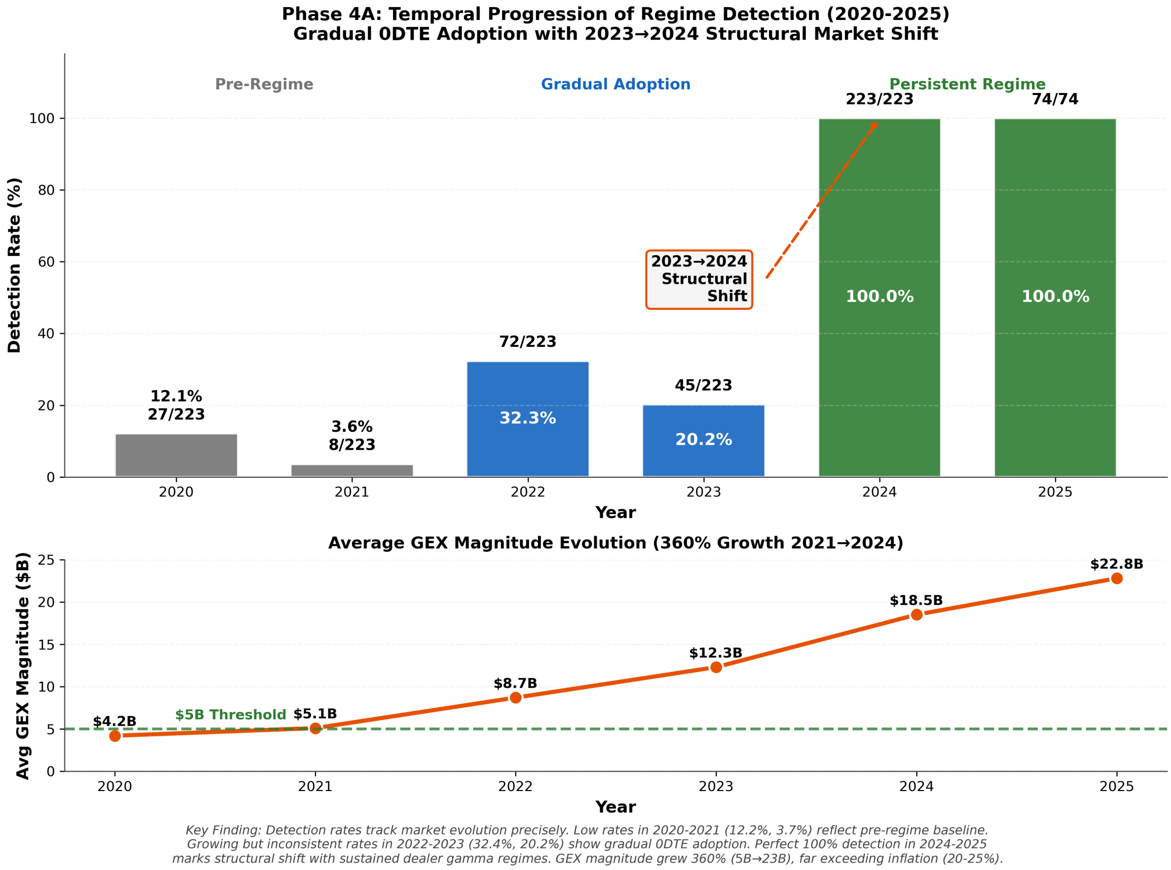


**Figure 8.** Temporal progression of regime detection (2020–2025). Detection rates evolved from pre-regime baseline through growing adoption to structural shift at 100% in 2024–2025. Average GEX magnitude grew from \$3.0B to \$20.3B, far exceeding inflation.

## 6. Discussion

### *6.1. From Single-Day to Multi-Day: Bridging Two Temporal Scales*

The progression from single-day detection (71.5%) to multi-day regime identification (81.2% in 2024) reveals complementary validation dimensions. Single-day testing establishes that the LLM understands dealer mechanics within individual trading sessions; regime detection validates that this understanding extends to persistent multi-day structural states. The raw chain finding (92.3% detection from first-principles data) provides additional support: if the model merely matched statistical signatures, raw chain performance would degrade rather than improve when pre-calculated summaries are removed.

The detection-alpha orthogonality finding from single-day analysis (Sharpe collapse from 1.8 to 0.1 while detection remains stable) motivated the regime extension: patterns that persist despite economic insignificance must reflect structural mechanics rather than exploitable inefficiencies. This distinction—mechanism identification versus alpha generation—is fundamental to deploying LLMs for market surveillance and risk management rather than trading.

### *6.2. Validating Structural Reasoning*

Five-phase validation provides evidence that the LLM performs structural reasoning rather than pattern matching. The 69.1 percentage point discrimination between 2024 (81.2%) and 2020 (12.1%) establishes selectivity: if the model relied on memorized associations or universal statistical patterns, detection rates would converge across periods rather than exhibiting 6.7× separation.

Three findings support this interpretation. First, the LLM maintained 98% mechanical accuracy across both detection conditions—correctly computing persistence, magnitude, and flip counts regardless of whether windows qualified as regimes. Second, negative controls achieved 0% false positives on transitional and low-magnitude synthetic data, confirming the model enforces structural criteria rather than over-detecting patterns. Third, confidence scores provide continuous quality discrimination beyond binary classification ($r = +0.719$ with persistence, $r = -0.705$ with sign flips), suggesting the model tracks regime robustness, not just threshold crossing.

This selectivity contrasts with earlier 5-day trajectory testing (Regan and Xie 2025), which achieved 98–100% detection across all market conditions by identifying universal daily hedging flows. The deliberate shift to 30-day windows with strict criteria transformed the task from universal pattern matching to selective regime identification—a critical distinction for validating genuine structural reasoning.

### *6.3. Market Structure Evolution and 0DTE Hypothesis*

The multi-year temporal analysis (Phase 5) reveals a gradual, non-monotonic evolution in regime detection that *coincides with* the adoption curve of zero-days-to-expiration (0DTE) options rather than a sharp structural break. Detection progressed from 12.2% (2020) through borderline years (3.7% in 2021, 32.4% in 2022) to sustained 100% detection in 2024–2025, with average GEX magnitude growing from \$3.0B to \$20.3B. We are careful to frame the 0DTE correspondence as temporal coincidence supported by a plausible mechanical channel (pinned daily dealer hedging demand), not as a demonstrated causal relationship; the observational design here cannot rule out alternative drivers, and we list those explicitly in §6.7.

The non-monotonic pattern (32.4% in 2022 dipping to 20.2% in 2023 before reaching 100% in 2024) is suggestive rather than definitive: 2023's elevated volatility (FOMC uncertainty, banking stress) appears to have disrupted regime formation despite growing 0DTE hedging pressure, which is consistent with the view that regime persistence requires both sustained dealer pressure *and* a volatility environment permitting consolidation. We read this dynamic as consistent with, rather than proof of, a 0DTE-mediated structural interpretation; stronger causal identification would require a natural experiment—for example a temporary 0DTE suspension, a 0DTE launch on a comparable non-SPY underlier that could serve as a counterfactual, or an instrumental-variable design that separates the 0DTE channel from contemporaneous shifts.

Concurrent factors that cannot be excluded in the observational data include: (i) the 2021–2023 interest-rate cycle ($\approx 0.25\% \rightarrow 5.5\%$), which reshaped the carry landscape for every option-writing strategy, (ii) the growth of systematic short-volatility flow into single-name and index options, (iii) increasing passive and index-linked assets under management, and (iv) changes in market-maker concentration following the 2020–2022 period of retail-driven volatility. Each of these could plausibly contribute to the detection pattern alongside 0DTE adoption. The non-monotonic detection trajectory is *less easily reconciled with* a simple gradual secular trend—detection remained low through 2023 despite continuous interest-rate increases and continuing technology diffusion, then jumped discontinuously in 2024 coincident with 0DTE market saturation ($\approx$46% SPY volume share)—but we emphasise that "less easily reconciled" is not "ruled out." Disentangling these channels is beyond the scope of this work, which focuses on LLM-validation methodology rather than on causal microstructure inference.

### *6.4. Temporal Obfuscation as Generalizable Methodology*

Our obfuscation testing framework generalizes beyond financial applications. The core principle—stripping temporal and contextual identifiers to force structural reasoning—addresses a universal challenge in deploying LLMs for domain-specific analysis: distinguishing genuine reasoning from training data contamination. The methodology requires three components: (1) identifiable features that can be systematically removed, (2) structural patterns that persist after obfuscation, and (3) negative controls validating discrimination.

Potential applications include medical time series analysis (removing patient identifiers and dates while preserving physiological patterns), climate data analysis (obfuscating location and temporal markers), and industrial anomaly detection (stripping equipment identifiers). In each case, temporal obfuscation could validate whether LLMs reason from structural properties or rely on memorized associations from training data.

### *6.5. Dispersed Knowledge and Information Aggregation*

The raw chain superiority finding (92.3% vs. 61.5%) admits a theoretical interpretation grounded in Hayek (1945)'s analysis of dispersed knowledge in economic systems. Hayek argued that the economic problem is fundamentally one of utilizing knowledge "not given to anyone in its totality"—that relevant information is distributed across countless individual participants, each possessing local knowledge inaccessible through centralized aggregation. Traditional parametric metrics like net GEX attempt precisely such centralized aggregation: compressing the full strike-level options surface into a single scalar value. Our results suggest this compression is lossy in a theoretically predictable way.

The raw options chain preserves what Hayek termed "knowledge of the particular circumstances of time and place"—the asymmetrical strike concentrations, implied volatility skew patterns, and spatial open interest distributions that reflect actual dealer positioning at each strike. When the LLM processes raw chain data, it functions as an information aggregator that synthesizes these dispersed local signals into a coherent structural assessment, identifying emergent order arising from countless decentralized dealer hedging constraints. The 30.8 percentage point improvement over GEX-assisted detection provides empirical evidence that this "local knowledge" embedded in individual strikes contains structural signal that scalar aggregation necessarily destroys.

This interpretation extends beyond methodological curiosity. The alpha disappearance puzzle—stable detection (68–74%) despite collapsing profitability (Sharpe 1.8 $\rightarrow$ 0.1)—parallels a core insight from Austrian capital theory: that structural knowledge of market coordination patterns differs fundamentally from actionable entrepreneurial knowledge (Kirzner 1973). Detecting that dealers are constrained to pro-cyclical hedging is structural knowledge; profiting from that detection requires entrepreneurial judgment about timing, magnitude, and competing market forces that our framework deliberately excludes. The orthogonality between detection and alpha is thus not a limitation but a feature: it confirms the framework identifies genuine structural mechanics rather than ephemeral trading opportunities.

### *6.6. Practical Implications*

The results carry concrete implications along three axes that matter most to financial practitioners: risk management, market efficiency, and the design of quantitative research pipelines. We address each in turn.

#### 6.6.1. Risk management

Our dealer-gamma regime detection is best understood as a *risk-regime indicator* rather than a trading signal. Because the framework achieves stable detection (68–74% quarterly) even as Sharpe ratios collapse from 1.8 to 0.1, its output is a reading of whether the market is currently operating in a mechanically constrained state—not a forecast of directional return. Three specific risk-management applications follow.

First, for **intraday volatility budgeting**, a persistent negative gamma regime implies amplified dealer chase-hedging and elevated realised volatility clustering on high-volume days; sell-side risk desks can use the 30-day regime classification as a leading indicator for volatility-of-volatility exposure and size gamma-exposure limits accordingly. Second, for **option-book hedging under OpEx concentration**, the known pinning dynamic around large-OI strikes (Ni et al. 2005) is substantially more forceful under positive-gamma regimes that we now classify explicitly; book hedgers may increase hedging frequency around OpEx when the framework detects a persistent positive regime.

Third, for **risk-scenario design**, 30-day regime history provides a natural conditioning variable for stress-test calibration: a 2020-style fragmented period and a 2024-style persistent negative period imply materially different joint distributions for realised volatility and spot-vol covariance.

#### 6.6.2. Market efficiency

The detection-alpha orthogonality result—stable structural detection that does not translate into exploitable profit—contributes to the ongoing debate about efficiency in optionized equity markets. Our evidence is consistent with a weakly efficient market in which *structural constraints* are reliably identifiable but already priced: arbitrageurs compete away the first-order profit from knowing a regime exists, yet the underlying mechanics that generate volatility clustering and OpEx pinning persist because they are mandatory, not opportunistic, actions by dealers. This reconciles two claims that are often treated as contradictory: that dealer-gamma positioning demonstrably influences short-horizon price dynamics (Anderegg and Sokolov 2022), and that systematic strategies exploiting that influence deteriorate as attention accumulates. The framework thus provides a positive account of why microstructure-aware research can be genuinely informative for risk without being genuinely informative for alpha.

#### 6.6.3. Practitioners: data-pipeline and model-deployment design

Two practitioner-facing design implications follow directly from the experimental results. First, the 30.8-percentage-point advantage of raw strike-level data over pre-aggregated GEX (92.3% versus 61.5% at the single-day scale) is evidence that common pipeline designs—which compress option chains into scalar summaries before analysis—are discarding signal that an LLM can reconstruct when given the raw input. This challenges the default of parametric aggregation and suggests that practitioners deploying LLM-based analysis should ingest granular data wherever feasible. The design principle generalises beyond options dealer flow to any domain where scalar metrics are conventional summary of distributional inputs: credit risk (single probabilities from granular exposure tapes), fixed-income surveillance (duration summaries from granular curve positions), and equity factor research (single factor loadings from granular return-attribution inputs) are the obvious candidates.

Second, the 0DTE-driven regime shift detected in 2022–2024 implies that static microstructure models calibrated to pre-2022 data will miss a structural reorganisation that our framework picks up. Practitioners running surveillance, risk, or execution models should treat the 2022–2024 period as a regime change requiring recalibration, not a drift-along-the-same-curve.

### *6.7. Limitations and Future Work*

Seven limitations merit explicit discussion, and we describe the specific future work each motivates.

**Single-asset scope**: All reported results concern SPY. SPY is deliberately chosen as the highest-liquidity and earliest 0DTE-enabled U.S. equity benchmark, but this choice leaves cross-asset generalization empirically untested. Dealer positioning in QQQ, IWM, single-name equities, and non-equity underliers (futures, rates, FX) may exhibit different regime dynamics because of differences in option chain depth, 0DTE availability, and the composition of end-users. Cross-asset replication is the single highest-priority item for future work; a pre-registered protocol applying the same obfuscation and regime-classification framework to at least one additional ETF (QQQ) and one individual equity (e.g., NVDA, AAPL) would directly test the transferability claim.

**Single-LLM dependence**: All 2,221 evaluations were produced by a single reasoning model (OpenAI o4-mini). The detection rates reported here are therefore conditional on this specific model's prior distribution over market-structure reasoning. Model-swap validation with alternative reasoning families (e.g., Anthropic Claude, OpenAI o3, Google Gemini, open-source reasoning models) using identical prompts and the same obfuscated sequences is a direct and low-cost extension. A cross-model agreement analysis would sharpen the distinction between the framework's structural-reasoning claim and any o4-mini-specific artefacts.

**Lack of independent external validation**: Our per-window ground-truth metrics (persistence, magnitude, sign flips) are computed from the same Alpha Vantage options feed used to construct the windows. We do not cross-validate detected regimes against an independent data source (CBOE DataShop, OPRA consolidated feed, or a commercial vendor such as SpotGamma or MenthorQ) or against an independent oracle of dealer positioning. External validation—both against a second options-data pipeline and against related microstructure observables (realised volatility, implied-realised spread, opening auction imbalance)—would strengthen the claim that the detected regimes correspond to a real, cross-verified phenomenon rather than an artefact of any single data provider.

**End-of-day measurement**: Our GEX_OI approach captures dealer inventory at the close but not intraday gamma dynamics; high-frequency flow data could refine detection, particularly for 0DTE contracts that expire within a single trading session. Intraday GEX surface reconstruction from streaming OPRA is a natural extension.

**Causal attribution**: The 0DTE hypothesis is supported by temporal coincidence and a theoretical mechanism but remains circumstantial; observational data cannot exclude alternative explanations such as post-pandemic monetary-policy shifts, passive-flow concentration, or market-maker inventory changes. A natural experiment—for instance a temporary 0DTE suspension or a regulatory halt during a market stress episode—would provide stronger causal evidence. We treat the 0DTE correspondence as consistent with our structural-regime detection rather than as a demonstrated causal channel (see §6.3).

**Shuffle test asymmetry**: The 61% false positive rate on 2024 shuffled data (versus 12.1% on 2020 shuffled data) reflects extreme regime persistence rather than framework failure—2024 regimes exhibit such dominant same-sign positioning that randomizing day order rarely disrupts the aggregate signal. This asymmetry is itself informative, confirming that 2024 regimes are defined by aggregate dominance rather than temporal sequencing, but it means the shuffle test's diagnostic power is lower in high-persistence regimes and should be interpreted accordingly.

**Threshold sensitivity**: All tested parameter configurations maintained substantial 2024-versus-2020 discrimination (see §5 sensitivity analysis), but the chosen thresholds (persistence $\geq$70%, magnitude $\geq$\$5B, flips $\leq$5) represent empirically validated design choices rather than first-principles derivations. Future work should explore adaptive thresholding that responds to volatility regime, contract-maturity mix, or prevailing options notional.

## 7. Conclusion

The primary contribution of this work is *methodological*: we presented temporal obfuscation testing as a generalizable procedure for validating LLM structural reasoning in domain-specific applications. Options dealer gamma-exposure regime detection served as the empirical demonstration domain—chosen because it combines theoretically grounded mechanical constraints, a large quantitative testbed, and a sharp pre-versus-post-0DTE temporal contrast—but the methodology is not specific to finance. The financial-market observations reported below are downstream evidence that the methodology discriminates persistent from fragmented structural regimes in ways consistent with known microstructure dynamics, and are not intended as novel claims about options market microstructure per se.

With that positioning in mind, comprehensive validation across 2,221 evaluations spanning 2020–2025 demonstrates four primary contributions:

1. **Single-day structural reasoning**: Temporal obfuscation achieves 71.5% detection of dealer hedging patterns with 91.2% predictive accuracy. Raw chain validation (92.3% vs. 61.5%) demonstrates that LLMs reconstruct dealer positioning from first principles, establishing that parametric GEX represents lossy compression of structural signal.

2. **Multi-day regime selectivity**: Extending to 30-day windows, the framework achieves 69.1 percentage point discrimination between persistent regimes (2024: 81.2%, 95% CI [75.8, 86.1]%) and fragmented markets (2020: 12.1%, 95% CI [8.1, 16.6]%; Fisher's exact $p = 1.8 \times 10^{-52}$, $\varphi = 0.69$), with Wilson upper bounds of 1.7%–10.7% on synthetic controls and 98% mechanical accuracy. A sensitivity sweep across 45 alternative threshold configurations confirms that the 2024-vs-2020 gap remains above 50 pp in 40 of 45 configurations, so the separation does not depend on the specific threshold choices. Unlike prior 5-day testing that achieved universal 98–100% detection, strict regime criteria produce selective 12–81% detection—validating structural state identification rather than universal pattern matching.
3. **Market structure evolution**: Multi-year analysis (1,412 windows, 2020–2025) reveals gradual regime evolution that coincides with 0DTE options adoption: detection progresses from 3.7% (2021) through a non-monotonic path to 100% (2024–2025), with average GEX magnitude growing from \$3.0B to \$20.3B. The non-monotonic path—requiring both sustained dealer pressure and volatility consolidation before regime detection stabilises—is consistent with, though does not on its own establish, 0DTE-driven structural reorganization. Alternative contemporaneous factors (interest-rate regime, passive-flow concentration, market-maker inventory changes) cannot be excluded with the observational data available here; stronger causal evidence would require a natural experiment such as a temporary 0DTE suspension.
4. **Detection-alpha orthogonality**: Stable detection (68–74% quarterly) despite collapsing economic profitability (Sharpe 1.8 $\rightarrow$ 0.1) confirms that detected patterns represent structural market mechanics rather than exploitable inefficiencies, establishing appropriate use cases in risk management and market surveillance.

**Future directions**: Three extensions emerge. First, cross-asset generalization to individual equities and other index ETFs would test whether regime detection reflects market-wide versus asset-specific dynamics. Second, extending temporal obfuscation to other domains (medical time series, climate data, industrial anomaly detection) would validate the methodology's generalizability beyond financial applications. Third, intraday GEX measurement incorporating 0DTE flow data may refine detection granularity, particularly for the ultra-short-maturity instruments driving modern market structure.

All code, data preprocessing scripts, and validation results are publicly available.[2]

## Appendix A Regime Detection LLM Prompt

This appendix reproduces the complete prompt submitted to the LLM for each 30-day regime-detection window, together with the API configuration and output schema, so that the experiment is fully reproducible from the published manuscript without reference to the source repository.

The authoritative implementation lives at `src/llm/mechanics_prompt_builder.py::build_regime_prompt()` in the code release accompanying this paper; the text below is transcribed verbatim from that implementation.

### *Appendix A.1 Model and API Configuration*

All 2,221 evaluations were obtained from OpenAI's `o4-mini` reasoning model through the OpenAI Batch API (asynchronous, 24-hour SLA, 100% completion rate observed across the five validation phases).

- **Model:** `o4-mini`
- **Temperature:** 1.0 (OpenAI reasoning models require a fixed temperature of 1; sampling-temperature adjustment is not exposed for `o1`, `o3`, or `o4` model families)

[2] https://github.com/iAmGiG/gex-llm-patterns

- **Maximum completion tokens:** 16,384
- **Response format:** JSON object (enforced via `response_format={"type":"json_object"}`)
- **Access mode:** OpenAI Batch API, batched 1,000 requests per submission

**Reproducibility note.** Because the reasoning models do not accept a user-supplied `seed` parameter and run at a fixed `temperature`, exact bit-identical replication of any single response is not guaranteed. Reproducibility at the *distributional* level is achieved by (i) the large sample size (N = 2,221 evaluations) and (ii) the mechanical criteria embedded in the prompt, which give the model concrete numerical thresholds to apply rather than asking for free-form judgment. Section 5 reports detection rates with bootstrap 95% confidence intervals to quantify the residual sampling variation.

*Appendix A.2 System Message and User Prompt*

The prompt is delivered as a single user-role message (the `o4-mini` reasoning model treats the first paragraph as the de facto system instruction). The placeholder `{gex_data_table}` is replaced at runtime with the obfuscated 30-day GEX sequence: one line per day, in the format `Day T-29:  +3.42B` through `Day T+0:  -12.18B`, where the calendar date has been replaced with a relative day label and the ticker symbol is absent. No other identifying context is supplied.

```
You are a market structure analyst specializing in dealer gamma
positioning regimes.

TASK: Analyze this 30-day period and determine if it represents a
PERSISTENT regime where dealer constraints create forced, directional
flows.

## 30-DAY GEX DATA

{gex_data_table}

## REGIME CLASSIFICATION FRAMEWORK

### PERSISTENT REGIMES (Detect These)

**1. PERSISTENT POSITIVE REGIME**
- Definition: Dealers are LONG gamma, forced to sell into strength
- Criteria:
  * >70% of days (21+/30) have positive net GEX
  * Average magnitude >$5B
  * <=5 sign flips across 30 days
  * Stable directional constraint

**Mechanism**: When dealers hold long gamma:
- Price rises -> Dealers MUST sell shares (rebalance)
- Price falls -> Dealers MUST buy shares (rebalance)
- Creates dampening, mean-reverting flows
- Constraint is STRUCTURAL (dealers cannot avoid)

**2. PERSISTENT NEGATIVE REGIME**
- Definition: Dealers are SHORT gamma, forced to buy into strength
- Criteria:
  * >70% of days (21+/30) have negative net GEX
```

```
  * Average magnitude >$5B
  * <=5 sign flips across 30 days
  * Stable directional constraint

**Mechanism**: When dealers hold short gamma:
- Price rises -> Dealers MUST buy shares (chase)
- Price falls -> Dealers MUST sell shares (chase)
- Creates amplifying, momentum flows
- Constraint is STRUCTURAL (dealers cannot avoid)

---

### NON-REGIMES (Reject These)

**3. TRANSITIONAL (Reject)**
- Frequent sign flips between positive/negative GEX
- No dominant regime direction (less than 70% same sign)
- Market in regime change period
- Example: 15 positive days, 15 negative days (50/50 split)

**Why Reject**: No persistent constraint. Dealers face mixed
conditions daily. Not a structural regime.

**4. LOW CONVICTION (Reject)**
- Consistent sign BUT weak magnitude (<$5B average)
- Example: 25 days positive, avg $2B GEX
- Insufficient constraint to create persistent forced flows

**Why Reject**: Even if sign is consistent, magnitude too weak to
force dealers into meaningful positions. Not a structural constraint.

---

## ANALYSIS QUESTIONS

Systematically evaluate the 30-day window:

**Step 1: Sign Persistence**
1. Count days with positive net GEX
2. Count days with negative net GEX
3. Calculate persistence percentage:
   max(positive_days, negative_days) / 30 * 100
4. Does it meet 70% threshold (21+ days)?

**Step 2: Magnitude Assessment**
1. Calculate average GEX magnitude (absolute value):
   sum(|net_gex|) / 30
2. Is average magnitude >=$5B?
3. Check for extreme outliers that might distort average
```

```
**Step 3: Stability Check**
1. Count sign flips: How many times does GEX switch from
   pos->neg or neg->pos?
2. Are there <=5 sign flips across 30 days?
3. Stable regime should have low flip count

**Step 4: Regime Classification**
- If Steps 1, 2, 3 all pass AND positive dominates
    -> PERSISTENT POSITIVE
- If Steps 1, 2, 3 all pass AND negative dominates
    -> PERSISTENT NEGATIVE
- If Step 1 passes but Step 2 fails -> LOW CONVICTION (reject)
- If Step 1 fails -> TRANSITIONAL (reject)

---

## CONFIDENCE CALIBRATION (Mechanical Guidance)

Use these concrete anchors to calibrate confidence:

**90-100 (Very High Confidence)**
- 25-30 days same sign (83-100% persistence)
- Average magnitude >$10B
- 0-2 sign flips (highly stable)
- Example: "29 negative days, avg $15B, 1 flip"

**70-89 (High Confidence)**
- 21-24 days same sign (70-80% persistence)
- Average magnitude $5-10B
- 2-4 sign flips (moderately stable)
- Example: "23 negative days, avg $7B, 3 flips"

**50-69 (Borderline - Use with Caution)**
- 18-20 days same sign (60-67% persistence)
- Average magnitude $3-5B
- 5-7 sign flips
- Example: "20 negative days, avg $4B, 6 flips"
- Note: Borderline cases should generally be REJECTED unless other
  factors strengthen confidence

**0-49 (Reject - Not Persistent)**
- <18 days same sign (<60% persistence)
- OR average magnitude <$3B
- OR >7 sign flips
- These are NOT persistent regimes

**Important**: Confidence is a FILTER, not a probability. Use it to
distinguish clear regimes (70+) from borderline (50-69) from noise
(<50).
```

```
---

## OUTPUT FORMAT (JSON)

Provide your analysis in this exact JSON structure:

{
    "regime_detected": true/false,
    "regime_type": "persistent_positive|persistent_negative|
                    transitional|low_conviction",
    "positive_days": <count as integer>,
    "negative_days": <count as integer>,
    "avg_magnitude_billions": <value as number>,
    "sign_flips": <count as integer>,
    "persistence_pct": <percentage as number>,
    "confidence": <integer 0-100>,
    "reasoning": "Explain step-by-step why this is/isn’t a persistent
                  regime. Reference specific metrics (persistence %,
                  avg magnitude, sign flips). If rejecting, state which
                  criterion failed."
}

**IMPORTANT**: All numeric fields (confidence, positive_days,
negative_days, sign_flips, avg_magnitude_billions, persistence_pct)
MUST be numbers (integers or decimals), NOT words like "thirty-five"
or "fifty".

**regime_detected Rules**:
- ‘true‘ ONLY if regime_type is "persistent_positive" or
  "persistent_negative"
- ‘false‘ if regime_type is "transitional" or "low_conviction"

---

## KEY PRINCIPLES

1. **Selectivity is Expected**: Most windows will NOT be persistent
   regimes (expect 30-50% detection rate)

2. **ALL Criteria Must Pass**: Persistence + Magnitude + Stability
   required for detection

3. **Rejection is Valid**: Saying "no persistent regime" is a correct
   answer for transitional/weak periods

4. **Mechanical Over Qualitative**: Use concrete thresholds
   (70%, $5B, 5 flips) rather than subjective judgment

5. **Structural Focus**: Only detect when dealers are FORCED into
   directional positions by constraints
```

```
Analyze the 30-day GEX data above and provide your regime
classification in JSON format.
```

*Appendix A.3 Output Schema and Parsing*

Each response is parsed into the following fields (types shown in parentheses):

- `regime_detected` (boolean) — `true` only when `regime_type` is `persistent_positive` or `persistent_negative`; `false` otherwise.
- `regime_type` (string) — one of `persistent_positive`, `persistent_negative`, `transitional`, `low_conviction`.
- `positive_days`, `negative_days`, `sign_flips` (integers in [0, 30]).
- `avg_magnitude_billions` (float, USD billions).
- `persistence_pct` (float, percentage).
- `confidence` (integer 0–100).
- `reasoning` (string) — free-form step-by-step explanation; retained for the post-hoc reasoning-quality audit reported in Section 5.

Parsing is performed by `src/validation/batch_regime_validator.py` via a robust JSON extractor that tolerates markdown code-fence wrappers and minor formatting drift. Any response failing schema validation is flagged for manual review; across the 2,221 evaluations in this study, the schema-validation failure rate was 0% (all responses were machine-parseable).

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## Abbreviations

The following abbreviations are used in this manuscript:

| | |
|---|---|
| GEX | Gamma Exposure |
| LLM | Large Language Model |
| 0DTE | Zero Days to Expiration |
| OI | Open Interest |

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